

Diligent Code Studio

User Manual - Version 0.6.0

Diligent Code Studio is a local-first development workbench for editing projects, checking dependencies, using AI assistance, managing Git status, running diagnostics, and preparing release packages.

What Changed in v0.6.0

Area	Behavior
AI Help	Minimized by default and opened only from the upper-right AI Help button.
AI output routing	Questions asked inside AI Help show the response inside AI Help. Full coding output stays in the AI Coding Assistant workspace page.
Editor screen	Removed the extra editor AI side panel so the editor keeps full width.
Manual button	The built-in PDF manual remains available from the upper-right Manual button.

Main Screen Areas

Workspace Menu: The top focus buttons move between Editor, AI Coding Assistant, Setup & Dependencies, Tool Registry, Terminal, Problems, Release Builder, Logs, Settings, and other tools. Drag the buttons to reorder them; reset the order from Settings.

Upper-right actions: Manual opens this PDF. Setup opens the dependency checklist. AI Help opens the compact navigation assistant.

AI Coding Assistant: Use this full page for larger coding answers, project summaries, README generation, installer scripts, bug finding, and code improvement recommendations.

AI Help vs. AI Coding Assistant

Diligent Code Studio intentionally separates quick navigation help from larger coding output. This prevents duplicate AI panels and keeps the workspace clean.

AI Help

AI Help is minimized by default. Open it from the upper-right AI Help button. Use it for quick questions like what the current screen does, where to find a feature, or what the next setup step should be. Responses appear inside the AI Help area.

AI Coding Assistant

Use the AI Coding Assistant workspace page for project-aware coding help. This is where full AI output appears for project review, bug finding, code improvement, missing file generation, README creation, installer scripts, and project summaries.

Recommended routing:

- Ask in AI Help -> response appears in AI Help.
- Ask in AI Coding Assistant -> response appears in the AI Coding Assistant window.
- Project-aware coding actions -> open/use the AI Coding Assistant window.

Project-Aware AI Actions

Ask AI About This Project: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Explain Current File: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Find Bugs: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Improve This Code: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Generate Missing File: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Create README: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Create Installer Script: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Summarize Project: Builds a focused prompt using available workspace context such as project type, active file, Git status, diagnostics, terminal output, and release state.

Setup & Dependencies

The Setup & Dependencies page checks whether required development tools are available on your computer. It can check and install common dependencies such as Git, GitHub CLI, Node.js, npm, Rust, Tauri tooling, Visual Studio Build Tools, WebView2 Runtime, Ollama, PowerShell, and Inno Setup.

If an installer reports that a package is already installed or no upgrade is available, Diligent Code Studio treats that result as success when the exit code and output indicate the tool is present.

Useful Commands

```
npm ci
npm run validate
npm run build
npm run tauri:dev
npm run tauri:build
```

GitHub Setup

Install Git and GitHub CLI from Setup & Dependencies, then sign in with GitHub CLI from Terminal or an external PowerShell window.

```
gh --version
gh auth login
gh auth status
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Testing, Security, and Release Workflow

Use the validation scripts before creating a release. The app includes smoke tests, unit checks, security checks for obvious secret patterns, TypeScript checking, and production build validation.

```
npm run validate
npm run test:security
npm run audit:npm
npm run check:rust
npm run test:rust
```

Use the Release Builder to run a clean frontend build, then a Tauri build, then package artifacts with SHA-256 checksums and release notes.

Troubleshooting

NPM window opens from installed app: Make sure you are launching the installed executable, not a dev script or npm command. v0.6.0 keeps startup checks manual and uses GUI-subsystem settings for the Windows app.

GitHub CLI installed but gh is not recognized: Close and reopen Diligent Code Studio and PowerShell so PATH refreshes. If needed, add C:\Program Files\GitHub CLI to PATH.

AI disabled: Open Settings and choose either Ollama for local use or OpenAI for API-based use. OpenAI API keys are session-only and not saved to preferences.

End of manual.